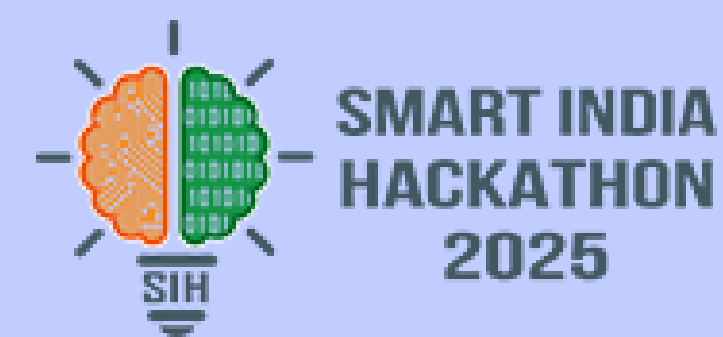
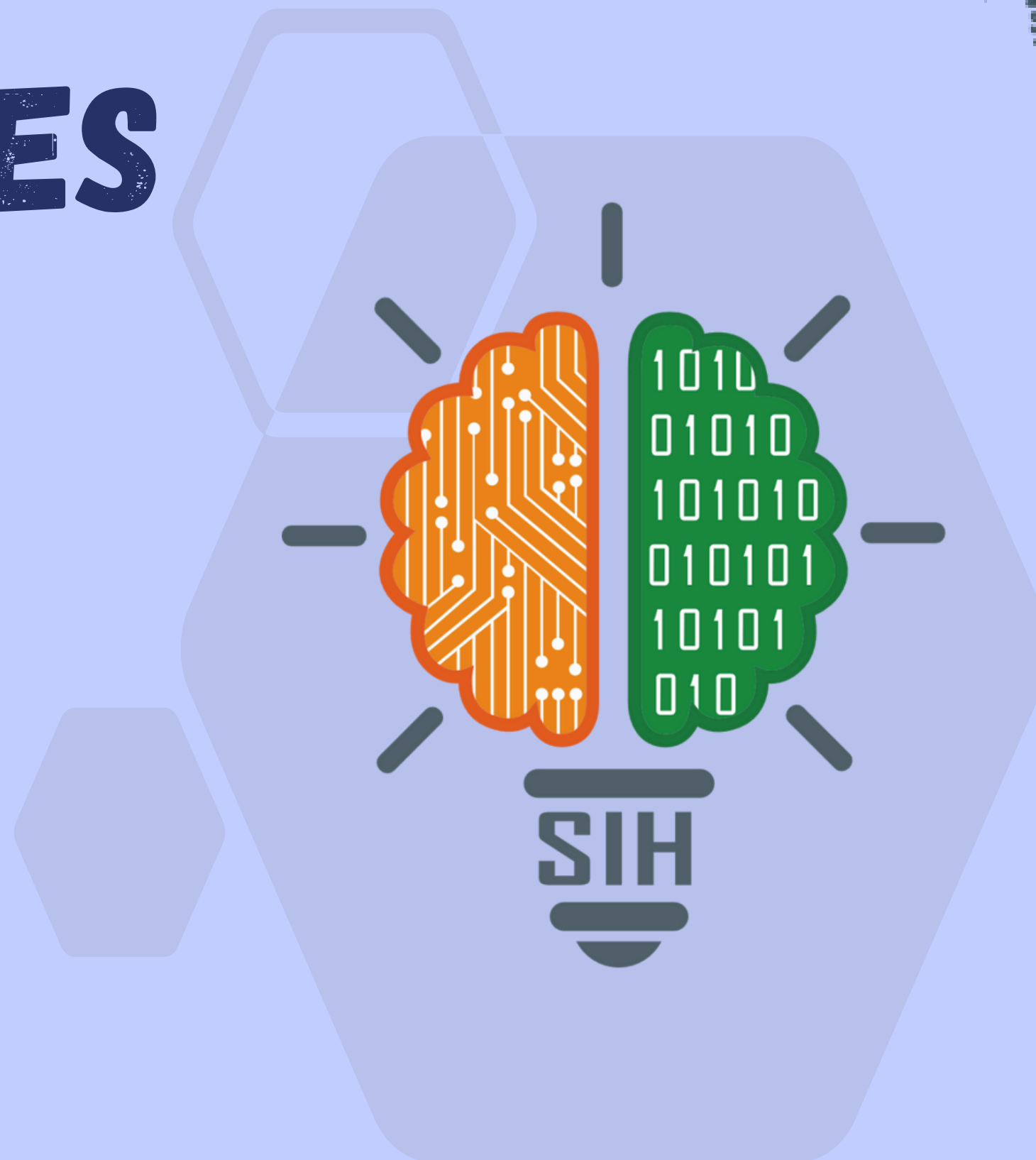


# SMART INDIA HACKATHON 2025



## DATABAES

- Problem Statement ID – 25031
- Problem Statement Title:  
  
Crowdsourced Civic Issue  
  
Reporting & Resolution System
- Theme-Clean and Green Technology
- PS Category- Software
- Team ID-
- Team Name (Registered on portal)



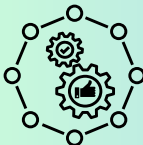
INNOVATION AND  
UNIQUENESS



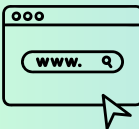
Revolutionary Spectator  
Role



Advanced Community  
Features



Smart Automation



Enhanced User  
Experience

PROBLEM SOLVED

Local governments face challenges in promptly identifying, prioritizing, and resolving civic issues due to lack of effective reporting mechanisms.



GPS-based real-time mobile reporting

Multi-factor scoring (urgency, clustering)

Automated routing + progress tracking

Dashboards, analytics & citizen feedback

Status tracking, upvoting & trending issues

SOLUTION OVERVIEW

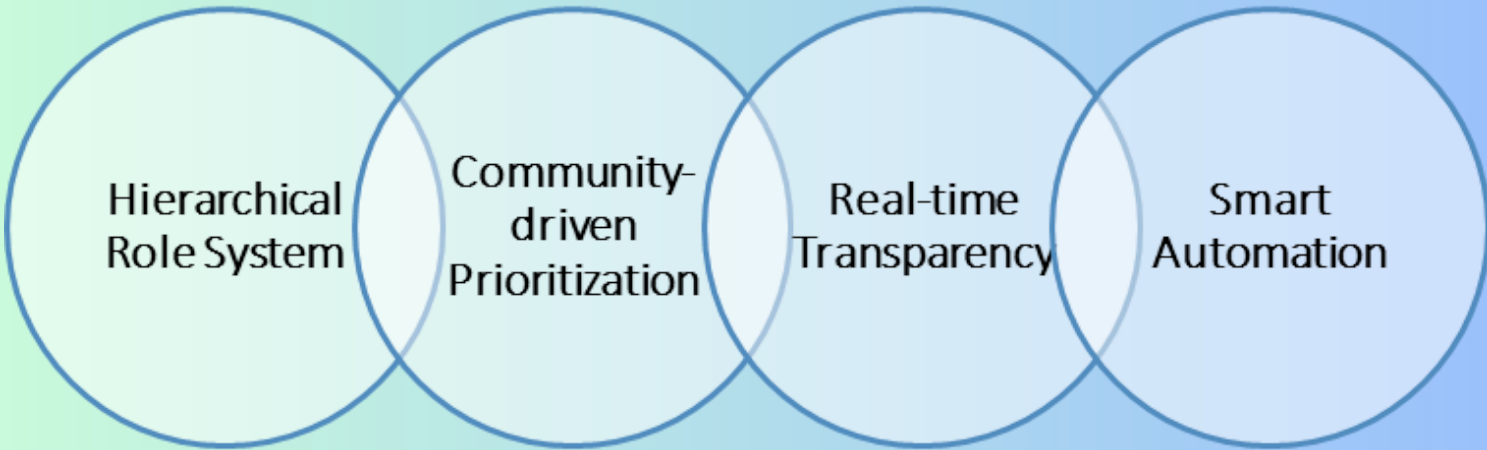
Three-Tier Architecture



Core Value Proposition



Key Differentiators



COMMON PROBLEMS IN CURRENT PLATFORMS

Low Adoption & Trust

Duplicate & Poor-Quality Reports

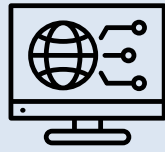
Wrong Department Routing

Inaccurate Location Data

Lack of Transparency

Accessibility & Digital Divide

## TECH STACK



### FRONTEND

- React Native
- React.js
- Tailwind CSS
- React Native Elements
- Google Maps API
- React Native Camera

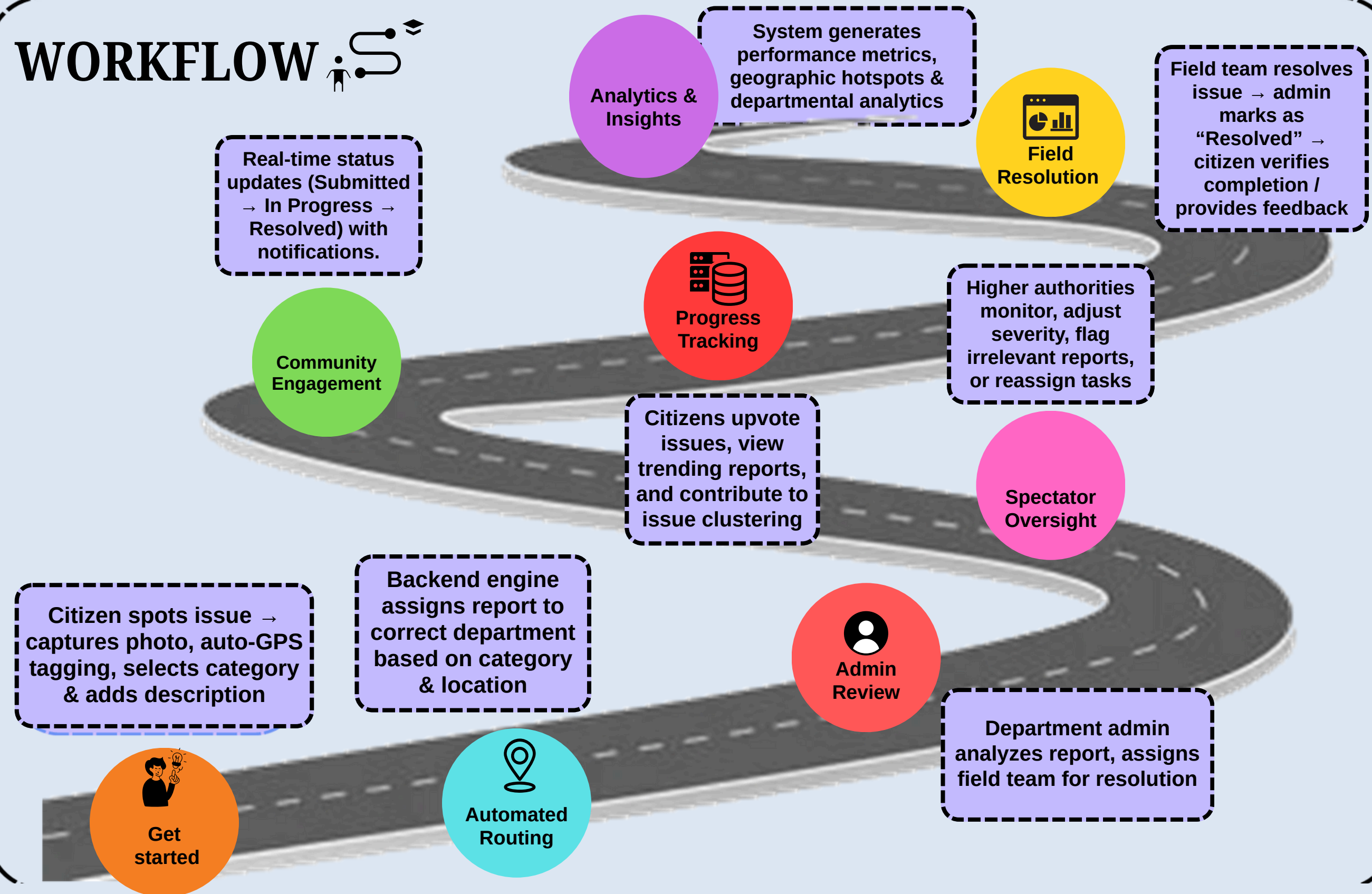
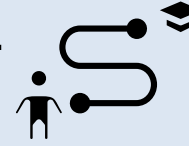
### BACKEND

- Node.js
- Express.js
- PostgreSQL
- Redis
- AWS S3
- JWT (JSON Web Tokens)
- Socket.io

### DEPLOYMENT TOOLS

- Git, GitHub
- GitHub Actions
- Jest, Cypress
- ESLint, Prettier
- Chart.js
- CloudFront

## WORKFLOW





## **FEASIBILITY ANALYSIS**

**Technical:** Built on React Native, Node.js, PostgreSQL, AWS S3; scalable, real-time, with Google Maps integration.

**Operational:** Mobile-first, multilingual, offline-ready; dashboards fit government workflows for easy adoption.

**Financial:** ₹15–25L development, ₹50K–1L monthly cloud; strong ROI via faster resolution & higher satisfaction.

**Legal:** IT Act 2000 compliant, open-source, meets govt. procurement & accessibility standards.

## **CURRENT CHALLENGES & RISKS**

**Technical:** Handling heavy media uploads, real-time sync, cross-platform parity, and legacy system integration.

**Operational:** Staff resistance, low digital literacy, and poor rural connectivity.

**Security:** Data breaches, spam/fake reports, and privacy concerns.

## **MITIGATION STRATEGIES**

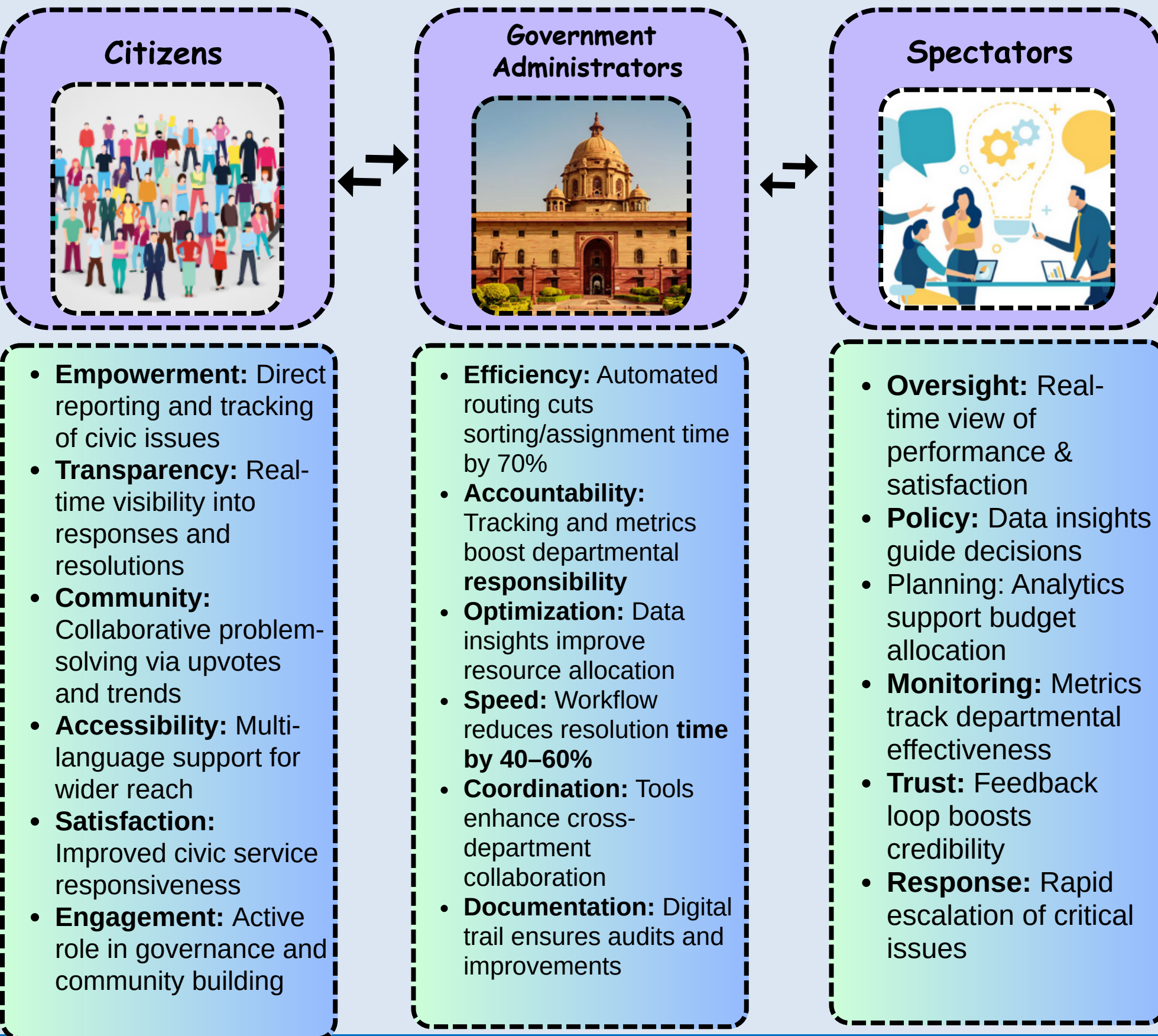
**Risk Mitigation:**

**Technical:** Auto-scaling, Redis caching, CDN compression, PWA fallback.

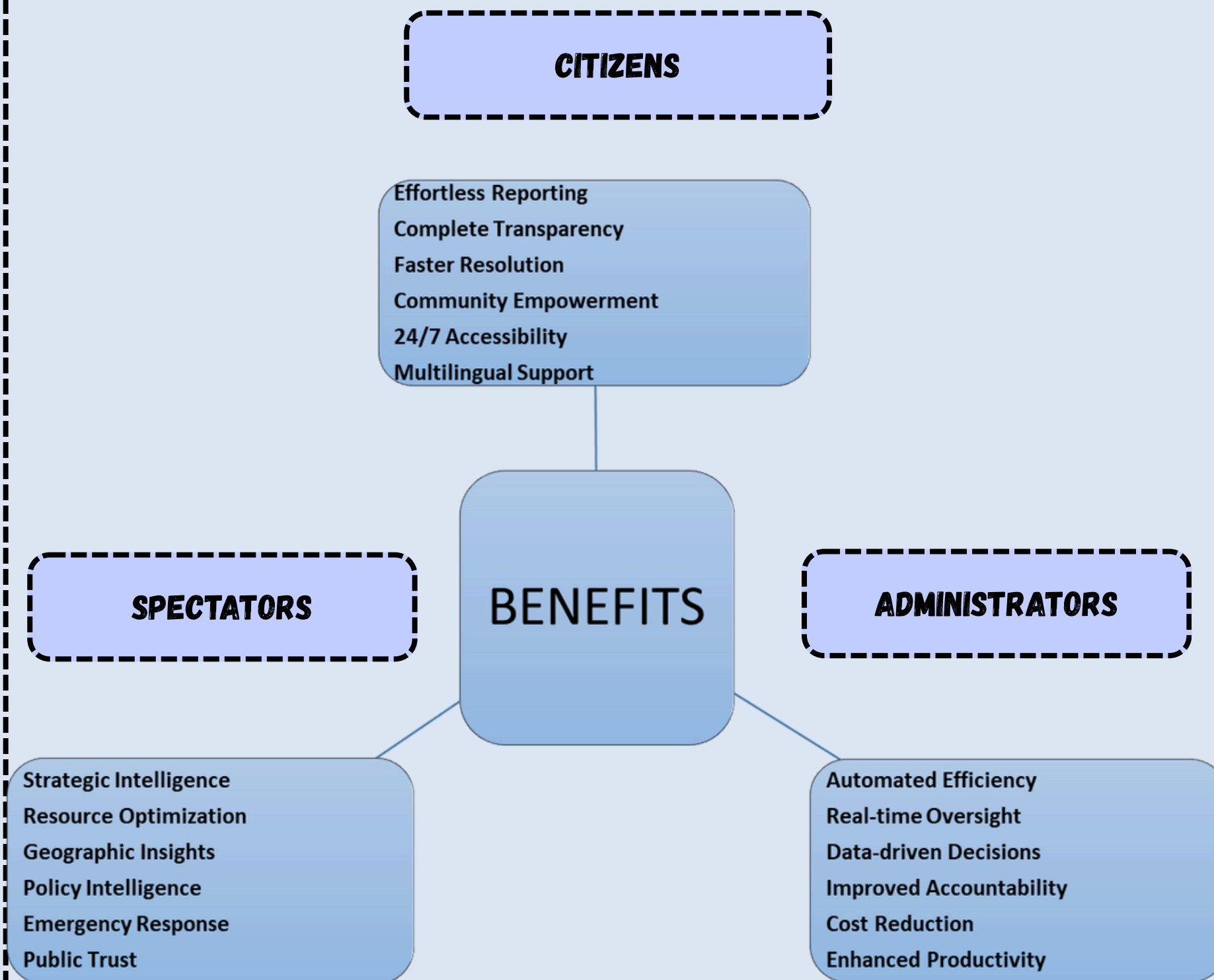
**Operational:** Staff training, NGO-led citizen education, SMS/USSD support, report quality checks.

**Security:** Encryption, rate limiting, privacy-by-design, regular audits & testing.

# POTENTIAL IMPACT ON TARGET



# BENEFITS



## ACADEMIC & INDUSTRY RESEARCH

- MIT Technology Review (2023): Digital governance and citizen engagement
- World Bank Report (2023): Mobile-first governance in emerging economies
- Journal of Urban Technology (2022): Crowdsourced urban problem-solving
- ACM Digital Gov Research (2023): Accessibility in civic apps



## TECHNOLOGY STANDARDS

- IT Act 2000 & PDP Bill: Indian data protection frameworks
- WCAG 2.1: Accessibility standards
- OWASP Mobile Security: Mobile app security best practices
- React Native Docs: Cross-platform development guides
- TensorFlow Lite: Lightweight ML for mobile inference

## GLOBAL CIVIC TECH STANDARDS

- Open311 Standard: RESTful civic issue reporting used in 20+ cities
- FixMyStreet (UK): 500K+ issues reported, 85% resolution rate
- SeeClickFix (USA): Used by 300+ municipalities, 2M+ issues resolved

## INDIAN GOVERNMENT INITIATIVES

- Digital India Programme: Framework for digital governance
- Swachh Bharat App: Cleanliness monitoring via citizen reporting
- PGRAMS: Public grievance redressal system
- Smart City Command Centers: Integrated urban management